

In 2009, India had the second highest number of GSoc participants—the proposals of 101 students were accepted.

## Cloud computing benefit

Perhaps the best way that Google has supported open source is by helping people move to Web applications and cloud computing. With applications like Google Mail, Google Calendar, Google Docs around, it really doesn't matter what operating system you use.

You may tell me that none of these are really open source. So what? If Linux does ever achieve a majority market share, do you think it won't have popular proprietary applications? At least most of Google's apps can be used via open source applications like Evolution, Thunderbird, Pidgin, etc. As these applications are all cross-platform, it makes switching from Windows to Linux a simple task. And since Google makes an effort to ensure all the data stored in its servers is easily accessible, there isn't any fear of data lock-in.

On May 27, 2009, when Google Wave—described by the company as a 'personal communication and collaboration tool'—was launched, everyone knew it was a potential game changer. The best thing about Google Wave is that Google plans to open source all the code. Hence, unlike Gmail, you can have your own Wave services on your own servers. So those who don't like using Gmail because of either privacy issues or its non-free nature, can use Wave without worries. Extensions can be easily developed to add new features to Wave. Also, since Wave is a tool for real-time communication and collaboration, it will help open source developers communicate much better.

And after the platform matures, developers can leave IRC channels and mailing lists, and switch to Wave. Some artists and documentation writers are already using Wave to collaborate better.

## The Android revolution

Of all open source software developed by Google, the one that has interested companies and individuals alike is the Android operating system. When Apple launched iPhone, users began to understand that the magic ingredient of a successful phone is its software. The iPhone helped create a demand for smart phones, and this meant other cellphone manufacturers had to look for a really good operating system for their phones.

Among Linux-based smart phone operating systems, Android is clearly the most successful. Google involved hardware developers in the creation of the Android, which was a good strategy. The Android has showcased Linux's scalability and adaptability in smaller form-factors. Android has also pushed the use of open source software in cellphones and helped put Linux in the hands of thousands who earlier depended on proprietary software.

## Chrome and the Chrome OS

The Chrome browser has managed to gain more than 4.5 per cent of the market to become the No. 3 browser in the world. But the real change that Chrome engineered was that it forced its competitors to produce better products. Chrome also brought with it an innovative and completely unique interface. Firefox 4 plans to include some of the ideas from Chrome's interface.

Google also developed a new algorithm called the Courgette, which helps provide faster and smaller updates for Chrome. Since this was also open sourced, many other projects considered using it for their update system. It is unfortunate, though, that Google was recently sued for patent infringement over Courgette, and its use is somewhat uncertain now. This brings us to the point of why Google is interested in creating its own open source operating system, namely, the Chrome OS.

Google's primary revenue channel is online advertising. It accounts for over 90 per cent of Google's income. For an organisation so heavily dependent on people clicking on Internet ads, it is natural for Google to try to get as many people online as possible. The two main factors that stand in the way are hardware and software. The hardware market is as competitive as it gets, with razor thin margins and products available to suit all kinds of budgets. But it is the software market which, because of the monopoly of a few players, becomes the bottleneck. One can easily see this reflected in the price of netbooks. If a netbook is available for Rs 12,000, then does it really make sense to pay a few more thousands for a copy of Windows?

This is precisely why Google decided to make its own OS and give it away for free. Many worry that this will only add to the problem of fragmentation in the open source ecosystem. But then, there are many reasons why they need not be afraid of the Chrome OS. The Chrome will be an OS on a secondary system. It will have only Web applications, mainly for netbooks. Another reason is that the features on the Chrome OS will be available through the Chrome browser. This means that whatever improvements Google makes in its Web applications or its OS, will be available to you. And even if a certain feature is not made available, you can easily port it, as it is all open source.

The Chrome OS brings with it some other advantages as well. OEMs like Dell and HP would feel more comfortable with an OS backed by Google, which can match Microsoft and Apple in popularity. Individuals, too, would rather try out an OS from Google than something not known to them. And in the end, since Chrome is all open source, and uses the Linux kernel, it will help improve other free operating systems as well. 

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